A complete fourteen-variable consistency equation

Strictly positive integer unknowns; exact degree 150; a 65-operation evaluator.

The only unknowns are

$$J, F, \alpha, z, f, i, T, s, w, n, \eta, \zeta, y, \gamma \in \mathbb{Z}_{>0}.$$

All other symbols in the equation are polynomial abbreviations:

$$\begin{gathered}
b = \beta J, \quad q = b + 1, \quad C = q - F - \alpha, \\
X = wq, \quad Y = sq^3, \quad E = XY, \quad k = \eta + \zeta, \\
\ell = EkY, \quad c = kY + \eta, \quad a = E + Y, \\
H = 4a + 3, \quad \Delta = a^2 + H, \quad D = X + ac + \gamma H, \\
R = [b(q - F) + \alpha](q^2 - 1) + (\mu + q\omega)J, \\
\tau = En + k - R, \quad V = f^2(cT - R) - c, \\
P = \tau^2 - \ell(\ell + k), \quad Q = D^2 - \Delta c^2, \\
S = i\Delta c^2, \quad U = S^2(V^2 - y^2) + y^2, \\
W = (K + w)C + (q - F) - zb, \quad Z = \Delta f^2 - S^2.
\end{gathered}$$

The single equation is

$$\boxed{PQUWZ - \Delta = 0.}$$

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\begin{gathered}
\rho = 2^{47}, \quad B = \rho^{2953}, \quad \beta = B - 1, \\
G_m(x) = \sum\nolimits_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{1945}.
\end{gathered}$$

$$\begin{aligned}
\mu &= \beta - G_{951}(\rho) \\
&\quad - \rho^{960} G_{26}(\rho) - \rho^{1005}, \\
\omega &= \beta + 4G_{917}(\rho) + (2^{45} - 4)\Lambda \\
&\quad + \rho^{960} G_9(\rho) + \rho^{987} G_8(\rho) + \rho^{1005}, \\
K &= \rho^{18} + \rho^{1005} + \rho^{2962} \\
&\quad + 2\Lambda\Phi(1/\rho),
\end{aligned}$$

where

$$\begin{aligned}
\Phi(x) = \sum\nolimits_{i=0}^{941} &[1 + 2^{10}(1 + w_i + 2^8 n_i \\
&+ 2^{17} p_i + 2^{25} s_i)]x^i \\
&+ 2^{10} x^{969}(x^9 + 2^{17}) G_8(2x) \\
&+ 2^{18} x^{942}(x^{18} + 2^{17}) G_9(2x).
\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 942 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$\begin{gathered}
h(q, b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\}, \\
r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q - 1\}|.
\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 941$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
0	$(2, 435, 3, 1)$
$j + 1$ $(j = 2q + b < 430)$	$((1 - d_j) r_{0,u_j}, h(q, b), d_j r_{1,u_j}, v_j)$
$433 - 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 432a + b)$
$\sum_{q \in \mathcal{S}_d,\ d,b=0,1}^{423 + 2r_{d,q} + b}$	$(dr_{d,q}, b, (1 - d) r_{d,q}, h(q, b))$
$923 + 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 438)$
927	$(2, 434, 2, 0)$
928	$(3, 436, 4, 4)$
929	$(4, 436, 4, 0)$
$496 + t$ $(t = 434, 435, 436)$	$(1, 438, 1, t)$
$933 + 3a + b$ $(a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A, k) = \lfloor A/b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$\begin{gathered}
\mathcal{N} = \mathcal{A} \bmod 2^{636}, \quad \mathcal{T} = \lfloor \mathcal{A}/2^{636} \rfloor \bmod 2^{1675}, \\
\mathcal{E} = \lfloor \mathcal{A}/2^{2311} \rfloor, \\
\nu_j = D_2(\mathcal{N}, j), \quad \sigma_j = \sum\nolimits_{k=0}^{j-1} \nu_k, \\
m_q = D_2(\mathcal{N}, 430 + q), \quad \tau_q = \sum\nolimits_{k=0}^{q-1} m_k, \\
F_q = 14 + m_q(D_{16}(\mathcal{E}, \tau_q) - 14), \\
u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(\mathcal{T}, j - \sigma_j), \\
v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).
\end{gathered}$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$ in base 10. Its complete decimal expansion is the concatenation of these lines:

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58607250878986781889126737426325463240251722983538552729956831
76496994940633391931532387237578247746672546301152856771829405
50763006124739855037353655204691391414186758883217097375044976
42946582709441420855053768425590291026930410855625406846907106
04079067372476312963058285538418194441497390173357046917308541
88716717442048170942517613524282407203789879020689779835988780
40029993694652746006732672118869311652887813524414818759918714
86564767549564149889156364118077528277278717925755937885232355
70139705060092827980034336115111037194604982010168556898406861
82116884452995397333290970898321098991845697808324750094293767
95970989537956914018656075189240808229394060280859606344247457
33090027075172177846970890324537666784074933609456239656992860
82694510289314542273549320441878654 9
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The ordinary complemented-color, native and Pell arguments give unsolvability iff Con(ZFC). Native calibration and internal-ZFC formalization remain open in Lean. This page is a complete finite specification, not an unsolvability proof.